SECTION 09670

RESIN BASED FLOORING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Carparks Resinous Flooring System

1.2 RELATED SECTIONS

A. Section 03300 - Cast-In-Place Concrete: Surface condition and concrete requirements.

1.3 REFERENCES

A. ASTM International (ASTM):
   A. ASTM D638 - Standard Test Method for Tensile Properties of Plastics.
   B. ASTM D645 - Standard Test Method for Thickness of Paper and Paperboard.
   C. ASTM D790 - Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials.
   D. ASTM D2240 - Standard Test Method for Rubber Property Durometer Hardness.
   E. ASTM D4060 - Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abrasion.
   F. ASTM D4541 - Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers.

1.4 SUBMITTALS

A. Submit under provisions of Section 01300.

B. Product Data: Manufacturer's data sheets on each product to be used, including:
   A. Manufacturer's product data.
   B. Manufacturer's installation instructions.

C. Test and Evaluation Reports: Certified test reports showing compliance with specified performance characteristics and physical properties.

D. Source Quality Control: Submit documentation verifying that components and materials specified in this Section are from single manufacturer.

E. Qualification Statements:
   A. Submit letter of verification for Manufacturer's Qualifications.
   B. Submit letter of verification for Installer's Qualifications.

F. Verification Samples: 12 inches by 12 inches (305 by 305 mm) samples of each resinous flooring system specified to show color and texture with specified coats cascaded.

1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications: Minimum 5 years’ experience manufacturing components similar to or exceeding requirements of project with sufficient capacity to produce and deliver required materials without causing delay in work, and capable of providing field service representation during construction.

B. Installer Qualifications: Minimum 2 years’ experience installing similar products and acceptable to the manufacturer.

C. Mock-Up: Provide a mock-up for evaluation of surface preparation techniques and application workmanship.
   A. Finish areas designated by Architect.
   B. Construct showing work.
   C. Dimensions and Process: Construct to 5 feet by 5 feet (1.52 m by 1.52 m) using proposed procedures, colors, textures, finishes and quality of work.
   D. Purpose: To judge quality of work, substrate preparation, operation of equipment and material application.
   E. Do not proceed with work prior to receipt of written acceptance of mock-up.
   F. When accepted, mock-up will demonstrate minimum standard of quality required for this work.
   G. Approved mock-up may remain part of finished work.
   H. Approved mock-up may not remain part of finished work. Remove mock-up and dispose of materials when no longer required.

D. Pre-installation Meetings: Conduct pre-installation meeting a minimum one week prior to commencing on-site installations to verify project requirements, substrate conditions and coordination with other building sub-trades, and to review manufacturer's installation instructions and manufacturer's warranty requirements.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Delivery and Acceptance Requirements: Deliver material in accordance with manufacturer's written instructions. Deliver materials in manufacturer's original packaging with identification labels intact and in sizes to suit project.

B. Storage and Handling Requirements: Store materials protected from exposure to harmful weather conditions and at temperature conditions recommended by manufacturer.

1.7 PROJECT CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's recommended limits.

B. Ambient Conditions:

A. Installation Location: Assemble and erect components only when temperatures are above 55 degrees F (13 degrees C).

B. Maintain materials, substrates and surrounding air temperature prior to, during, and 48 hours after completion to comply with product requirements.

PART 2 PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturer: Saudi Vetonit Company Ltd.-Saveto
71 St. Second Industrial Estate Riyadh P.O.Box 52235 Riyadh 11563 Saudi Arabia Tel +966 11 2653334, Fax: +966 11 265 0936,
E-mail: info.Riyadh@saveto.com; Web: www.saveto.com

B. Substitutions: Not permitted.

C. Requests for substitutions will be considered in accordance with provisions of Section 01600.

2.2 Materials :

2.2.1 Exposed Roof Deck & Waterproofing system

A. Primer: Vetoprime EP490: Versatile Solvent based epoxy primer as manufactured by Saudi Vetonit Co. Ltd.-Saveto to achieve 1200 μ Dry film thickness.(DFT).

Description:

1. Mixed Density:	~1.0 Kg / Liter
2. Solids Content (Volume):	50%
3. Application Thickness:	100 μ
4. Appearance:	Transparent Yellowish Liquid
5. Adhesion Bond to concrete:	Higher than host concrete cohesive strength.

A. Intermediate Coats consists of two coats of Vetotop UC363: High performance, aromatic & flexible waterproof polyurethane flooring membrane to achieve 800 μ per coat.

Description:

Specific Gravity:	1.30 ± 0.05
Solid Content (%Volume):	100
Tensile Strength (ASTM D638-14:	8 MPa
Elongation (ASTM D522/D522M-13):	80%
Bond Strength to Concrete ASTM D4541-09e1:	> 1.5 MPa
Taber Abrasion ASTM D4060 CS17 Wheels (mg loss/1000cycles)	55
Water Absorption ASTM D570-98(2010) e1:	0.001
Porosity with no sealer NACE Sand TM-01-74:	0

A. Anti-Slip aggregates : Vetograin Range Silica hardwearing aggregates to be used in conjunction of Saveto Anti-slip flooring system:

Description:

Vetograin Range	Vetograin size	Floor Texture
Vetograin 0.45	0.10-0.45mm	Extra fine
Vetograin 0.60	0.30-0.60 mm	Fine
Vetograin 0.80	0.425-0.80 mm	Medium
Vetograin 2.00	1.0-2.0 mm	Coarse

B. Vetotop UC371: Two component solvent base UV resistant Polyurethane offers excellent scratch resistant, achieving a dry fil thickness (DFT) of 100 µ

Description

1. Solid content (%100):	50
2. Bond Strength to Concrete ASTM D4541:	>1.5 MPa.
3. Taber Abrasion ASTM D4060 CS17 Wheels (mg loss/1000cycles):	20
4. Water Absorption ASTM D413 (maximum):	0.001
5. Porosity with no sealer NACE Sand TM-01-74:	0
6. Skid Resistance ASTM D2394:	Pass
7. Application thickness (DFT):	100 µ

2.2.2 External Ramps

One prime coat & two coats of Polyurethane coating Vetotop UC363 Incorporating silica sand in between to achieve anti-skid surface and top coat of Vetotop UC371 to achieve a total dry thickness of 1200 µ.

C. Primer: Vetoprime EP490: Versatile Solvent based epoxy primer as manufactured by Saudi Vetonit Co. Ltd.-Saveto

Description:

1. Mixed Density:	~1.0 Kg / Liter
2. Solids Content (Volume):	50%
3. Application Thickness:	100 µ
5. Appearance:	Transparent Yellowish Liquid
6. Adhesion Bond to concrete:	Higher than host concrete cohesive strength.

D. Intermediate Coats consists of two coats of Vetotop UC363: High performance, aromatic & flexible waterproof polyurethane flooring membrane to achieve 300 µ per coat.

Description:

1. Specific Gravity:	1.30 ± 0.05
2. Solid Content (%Volume):	100
3. Tensile Strength (ASTM D638-14:	8 MPa
4. Elongation (ASTM D522/D522M-13):	80%
5. Bond Strength to Concrete ASTM D4541-09e1:	> 1.5 MPa
6. Taber Abrasion ASTM D4060 CS17 Wheels (mg loss/1000cycles)	55
7. Water Absorption ASTM D570-98(2010) e1:	0.001
8. Porosity with no sealer NACE Sand TM-01-74:	0

E. Anti-Slip aggregates : Vetograin Range Silica hardwearing aggregates to be used in conjunction of Saveto Anti-slip flooring system:

Description:

Vetograin Range	Vetograin size	Floor Texture
Vetograin 0.45	0.10-0.45mm	Extra fine
Vetograin 0.60	0.30-0.60 mm	Fine
Vetograin 0.80	0.425-0.80 mm	Medium
Vetograin 2.00	1.0-2.0 mm	Coarse

F. Vetotop UC371: Two component solvent base UV resistant Polyurethane offers excellent scratch resistant, achieving a dry fil thickness (DFT) of 100 µ

Description

1. Solid content (%100):	50
2. Bond Strength to Concrete ASTM D4541:	>1.5 MPa.
3. Taber Abrasion ASTM D4060 CS17 Wheels (mg loss/1000cycles):	20
4. Water Absorption ASTM D413 (maximum):	0.001
5. Porosity with no sealer NACE Sand TM-01-74:	0
6. Skid Resistance ASTM D2394:	Pass
7. Application thickness (DFT):	100 µ

PART 3 EXECUTION

3.1 EXAMINATION

A. Do not begin installation until substrates have been properly prepared.

B. If substrate preparation is the responsibility of another installer, notify Architect of unsatisfactory preparation before proceeding.

3.2 PREPARATION

A. Clean surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

3.3 INSTALLATION

A. Install in accordance with manufacturer's instructions and in proper relationship with adjacent construction.

3.4 PROTECTION

A. Protect installed products until completion of project.

B. Touch-up, repair or replace damaged products before Substantial Completion.

END OF SECTION